

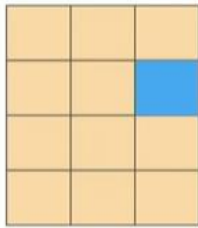
Module 7: Storage

1. AWS EBS

AWS storage options: Block storage versus object storage

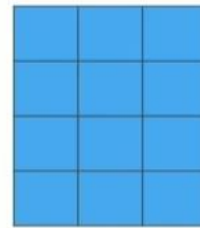


What if you want to change one character in a 1-GB file?



Block storage

Change one block (piece of the file) that contains the character



Object storage

Entire file must be updated

Amazon EBS

Amazon EBS enables you to **create individual storage volumes** and **attach them** to an Amazon EC2 instance:

- Amazon EBS offers block-level storage.
- Volumes are automatically replicated within its Availability Zone.
- It can be backed up automatically to Amazon S3 through snapshots.
- Uses include –
 - Boot volumes and storage for Amazon Elastic Compute Cloud (Amazon EC2) instances
 - Data storage with a file system
 - Database hosts
 - Enterprise applications

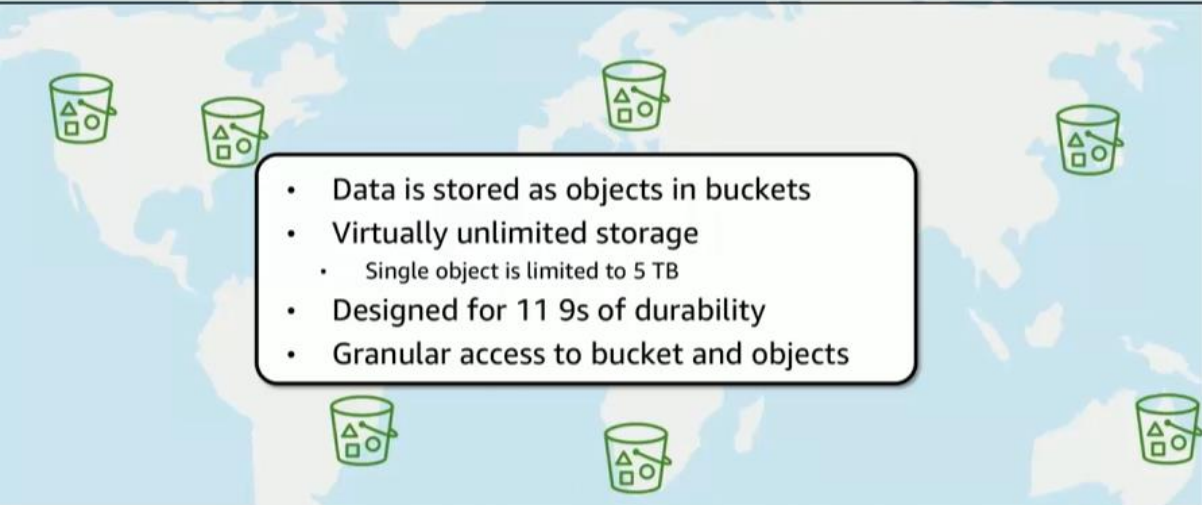
Amazon EBS features

- Snapshots –
 - Point-in-time snapshots
 - Recreate a new volume at any time
- Encryption –
 - Encrypted Amazon EBS volumes
 - No additional cost
- Elasticity –
 - Increase capacity
 - Change to different types

2. AWS S3

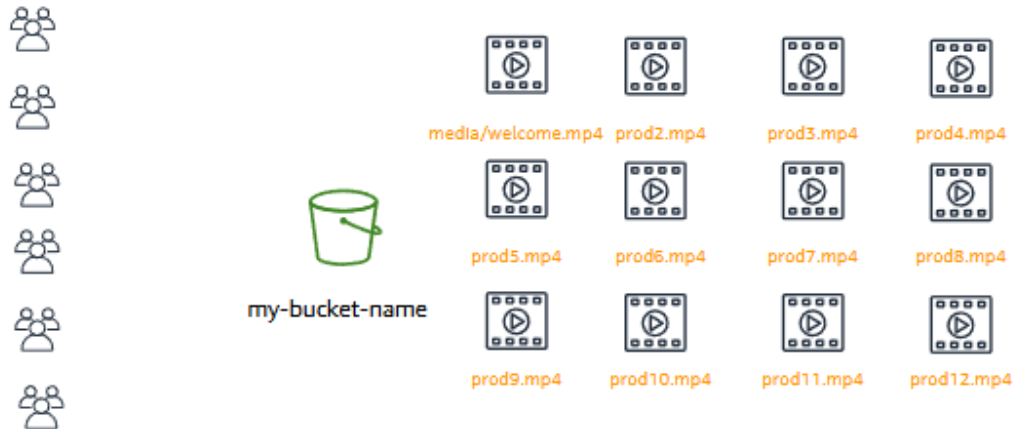
Amazon S3 overview

aws academy



- Data is stored as objects in buckets
- Virtually unlimited storage
 - Single object is limited to 5 TB
- Designed for 11 9s of durability
- Granular access to bucket and objects

Designed for seamless scaling



Access the data anywhere



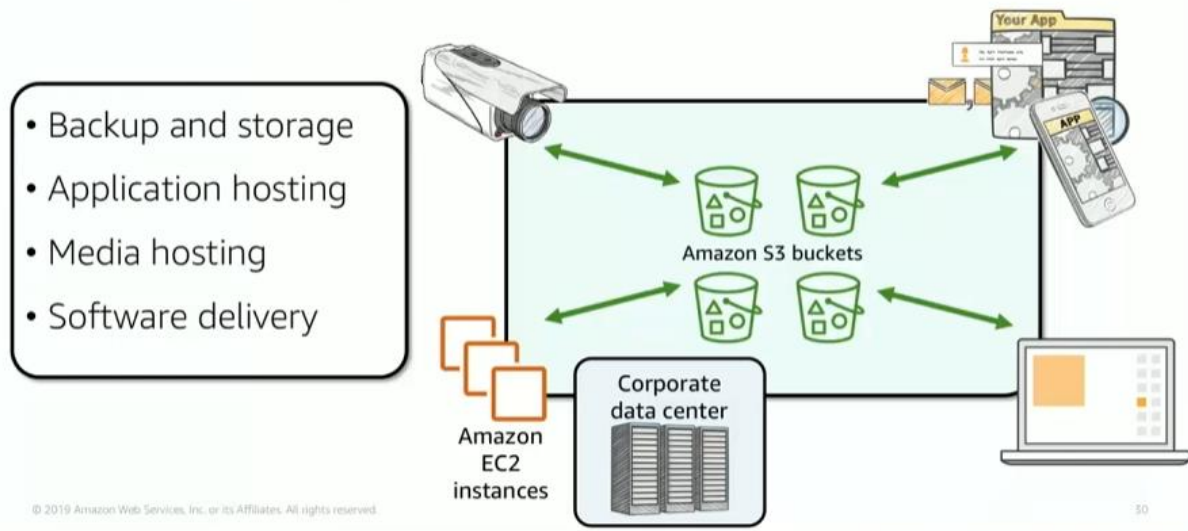
AWS Management
Console



AWS Command Line
Interface



SDK



3. AWS EFS

Amazon Elastic File System (Amazon EFS) provides simple, scalable, elastic file storage for use with AWS services and on-premises resources. It offers a simple interface that enables you to create and configure file systems quickly and easily.

Amazon EFS features

- File storage in the AWS Cloud
- Works well for big data and analytics, media processing workflows, content management, web serving, and home directories
- Petabyte-scale, low-latency file system
- Shared storage
- Elastic capacity
- Supports Network File System (NFS) versions 4.0 and 4.1 (NFSv4)
- Compatible with all Linux-based AMIs for Amazon EC2

- ① Create your Amazon EC2 resources and launch your Amazon EC2 instance.
- ② Create your Amazon EFS file system.
- ③ Create your mount targets in the appropriate subnets.
- ④ Connect your Amazon EC2 instances to the mount targets.
- ⑤ Verify the resources and protection of your AWS account.

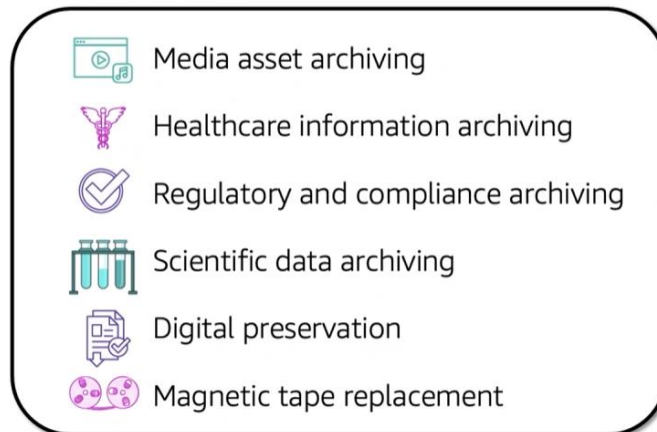
4. AWS S3 Glacier

Amazon S3 Glacier review

Amazon S3 Glacier is a **data archiving service** that is designed for **security, durability**, and an **extremely low cost**.

- Amazon S3 Glacier is designed to provide 11 9s of durability for objects.
- It supports the encryption of data in transit and at rest through Secure Sockets Layer (SSL) or Transport Layer Security (TLS).
- The Vault Lock feature enforces compliance through a policy.
- Extremely low-cost design works well for long-term archiving.
 - Provides three options for access to archives—expedited, standard, and bulk—retrieval times range from a few minutes to several hours.

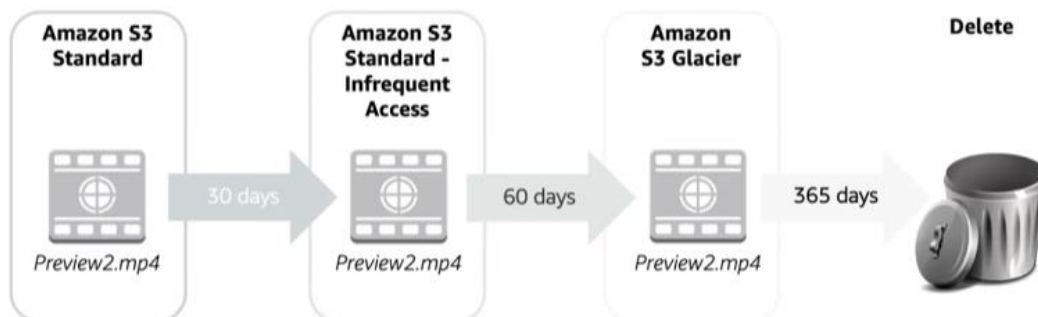
Amazon S3 Glacier use cases



Lifecycle policies



Amazon S3 lifecycle policies enable you to delete or move objects based on age.



Security with Amazon S3 Glacier



Amazon S3 Glacier



Control access with IAM



Amazon S3 Glacier encrypts your data with AES-256



Amazon S3 Glacier manages your keys for you